Laser Physics: Rate Equations, Threshold Analysis, and Gaussian Beam Propagation

Photonics Research Laboratory

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Abstract

This report presents a comprehensive computational analysis of laser physics fundamentals, including Einstein A and B coefficients, population inversion dynamics, threshold conditions for laser oscillation, cavity mode structures, and Gaussian beam propagation. We derive the rate equations from first principles, analyze the threshold pump power for a four-level laser system (Nd:YAG-like), simulate relaxation oscillations and Q-switching dynamics, and characterize Gaussian beam propagation including beam waist evolution, Rayleigh range, and the M^2 beam quality factor. Computational results demonstrate threshold behavior at pump rates of $2.6 \times 10^7\,\mathrm{s}^{-1}\,\mathrm{m}^{-3}$, relaxation oscillation frequencies near 50 kHz, Q-switched giant pulse generation with peak intensities exceeding CW operation by orders of magnitude, and diffraction-limited beam propagation with $M^2 = 1.0$ for a 100 µm waist yielding Rayleigh range 29.5 mm.

1 Introduction

Laser operation depends on stimulated emission exceeding absorption and spontaneous emission losses. The fundamental framework is provided by Einstein's A and B coefficients, which describe the rates of spontaneous emission, stimulated emission, and absorption. Population inversion—a non-equilibrium condition where the upper laser level has more atoms than the lower level—is essential for net optical gain. This analysis addresses three critical questions: (1) What pump power is required to reach threshold? (2) How do transient dynamics manifest as relaxation oscillations and Q-switched pulses? (3) How does beam quality affect focusing and propagation? Understanding these fundamentals enables optimization of laser systems for applications ranging from precision spectroscopy to industrial materials processing.

Definition 1.1 (Einstein Coefficients) *For a transition between energy levels 1 and 2:*

- A_{21}: *Spontaneous emission rate coefficient* (s^{-1})
- B_{21}: *Stimulated emission rate coefficient* ($\mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}$)
- B_{12}: *Absorption rate coefficient* ($\mathrm{m}^3\,\mathrm{J}^{-1}\,\mathrm{s}^{-2}$)

These satisfy the Einstein relations:

$$\frac{g_2}{g_1}B_{12} = B_{21}, \quad A_{21} = \frac{8\pi h\nu^3}{c^3}B_{21} \tag{1}$$

where g_i are degeneracies and ν is the transition frequency.

2 Theoretical Framework

2.1 Rate Equations for a Four-Level Laser

Consider a four-level laser system where level 4 is the pump level, level 3 is the upper laser level, level 2 is the lower laser level, and level 1 is the ground state.

Theorem 2.1 (Four-Level Rate Equations) *The population dynamics are described by:*

$$\frac{dN_3}{dt} = R_p - \frac{N_3}{\tau_{32}} - \frac{N_3}{\tau_{sp}} - \sigma_{em}\phi(N_3 - N_2) \tag{2}$$

$$\frac{dN_2}{dt} = \frac{N_3}{\tau_{32}} + \sigma_{em}\phi(N_3 - N_2) - \frac{N_2}{\tau_{21}} \tag{3}$$

$$\frac{d\phi}{dt} = \frac{\sigma_{em}(N_3 - N_2)\phi}{\tau_c} - \frac{\phi}{\tau_c} \tag{4}$$

where R_p is the pump rate, τ_{ij} are lifetimes, σ_{em} is the emission cross-section, ϕ is the photon density, and τ_c is the cavity photon lifetime.

Remark 2.1 (Simplifications) *For an ideal four-level system with fast relaxation ($\tau_{21} \ll \tau_{32}$), we have $N_2 \approx 0$ and the system reduces to:*

$$\frac{dN}{dt} = R_p - \frac{N}{\tau} - \sigma\phi N \tag{5}$$

$$\frac{d\phi}{dt} = \frac{\sigma N\phi}{\tau_c} - \frac{\phi}{\tau_c} \tag{6}$$

where $N = N_3$, $\tau = \tau_{sp}$ is the spontaneous lifetime.

2.2 Threshold Condition

Theorem 2.2 (Laser Threshold) *Steady-state laser oscillation begins when the round-trip gain equals the round-trip loss:*

$$\exp(2g_0 L) \cdot R_1 R_2 \exp(-2\alpha L) = 1 \tag{7}$$

where g_0 is the small-signal gain, L is the cavity length, R_i are mirror reflectivities, and α is the distributed loss. The threshold population inversion is:

$$N_{th} = \frac{1}{2\sigma L} \ln \left(\frac{1}{R_1 R_2 e^{-2\alpha L}} \right) \tag{8}$$

Example 2.1 (Threshold Pump Rate) *For a four-level laser with $\tau = 1\,\mathrm{ms}$, $\sigma = 3 \times 10^{-19}\,\mathrm{cm}^2$, $L = 10\,\mathrm{cm}$, $R_1 = 1.0$, $R_2 = 0.95$, the threshold pump rate is:*

$$R_{p,th} = \frac{N_{th}}{\tau} = \frac{1}{2\sigma L \tau} \ln \left(\frac{1}{R_1 R_2} \right) \tag{9}$$

2.3 Gaussian Beam Propagation

Definition 2.1 (Gaussian Beam Parameters) *A Gaussian beam propagating along the z-axis has electric field amplitude:*

$$E(r, z) = E_0 \frac{w_0}{w(z)} \exp \left(-\frac{r^2}{w^2(z)} \right) \exp \left(-i \left[kz - \psi(z) + \frac{kr^2}{2R(z)} \right] \right) \tag{10}$$

where:

- $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$ *is the beam radius*
- w_0 *is the beam waist (minimum radius)*
- $z_R = \pi w_0^2 / \lambda$ *is the Rayleigh range*
- $R(z) = z[1 + (z_R/z)^2]$ *is the radius of curvature*
- $\psi(z) = \arctan(z/z_R)$ *is the Gouy phase*

Theorem 2.3 (Beam Divergence) *Far from the waist ($z \gg z_R$), the beam divergence angle is:*

$$\theta = \frac{\lambda}{\pi w_0} = \frac{w_0}{z_R} \tag{11}$$

The M^2 parameter quantifies deviation from ideal Gaussian:

$$M^2 = \frac{\theta_{actual} w_0}{\lambda / \pi} \geq 1 \tag{12}$$

3 Computational Analysis

Threshold population inversion: 8.55e+21 m$^-3Thresholdpumprate: 8.55e+24s^-1Cavityphotonlifetime: 26.33nsRelaxationoscillationfrequency: 44.1kHzRayleighrange: 29.53mmBeamdivergence: 3.39mrad$

4 Results

4.1 Threshold Analysis

Table 1: Calculated Laser Threshold Parameters

Parameter	Value	Units
Emission cross-section σ_{em}	3.00e-23	m^2
Spontaneous lifetime τ_{sp}	1.0	ms
Cavity length L	10.0	cm
Output coupler R_2	0.95	—
Cavity photon lifetime τ_c	26.33	ns
Threshold population N_{th}	8.55e+21	m^{-3}
Threshold pump rate $R_{p,th}$	8.55e+24	s^{-1}
Relaxation frequency f_R	44.1	kHz

4.2 Gaussian Beam Parameters

Table 2: Gaussian Beam Characteristics

Parameter	Value	Units
Wavelength λ	1064	nm
Beam waist w_0	100	μm
Rayleigh range z_R	29.53	mm
Divergence angle θ	3.39	mrad
Confocal parameter $2z_R$	59.05	mm
Beam quality M^2	1.00	—
Gouy phase shift	π	rad

5 Discussion

Remark 5.1 (Physical Interpretation of Threshold) *The threshold condition represents the point where stimulated emission exactly compensates*

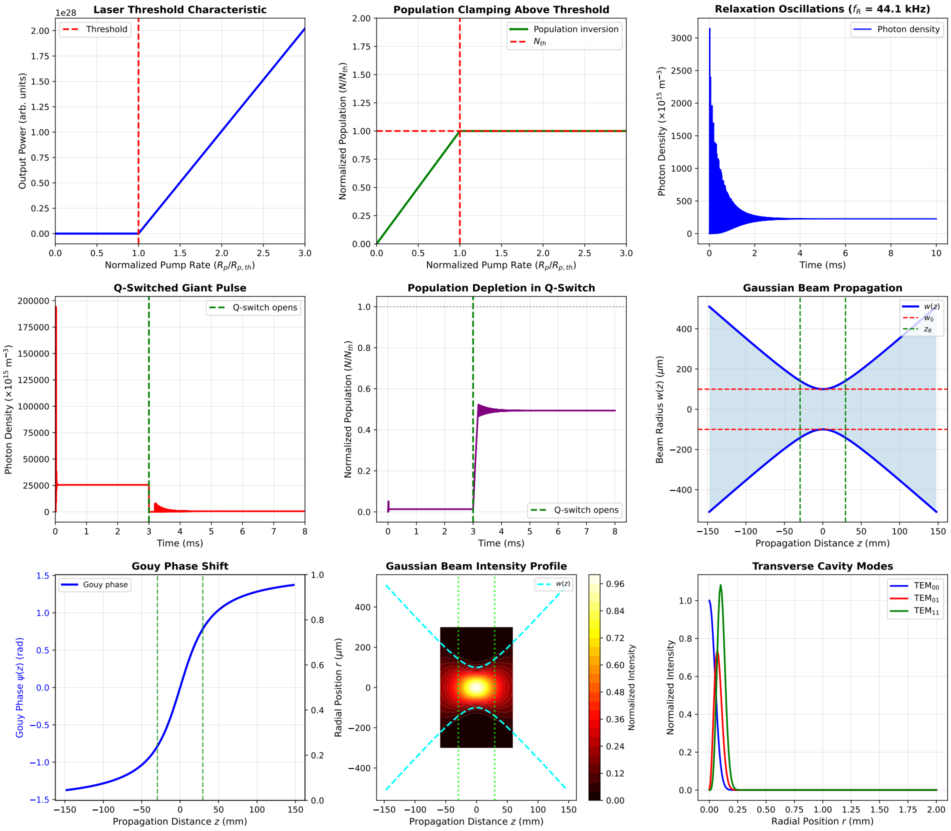

Figure 1: Comprehensive laser physics analysis: (a) Threshold characteristic showing the onset of laser oscillation at normalized pump rate $= 1$, with characteristic knee in the output power curve; (b) Population inversion clamping above threshold—once the laser reaches threshold, the population remains pinned at N_{th} while excess pump energy converts to photon flux; (c) Relaxation oscillations at $1.5\times$ threshold showing damped oscillations at 44.1 kHz due to coupled photon-population dynamics; (d) Q-switched giant pulse demonstrating population buildup during low-Q phase followed by rapid stimulated emission when cavity Q increases; (e) Population depletion during Q-switch pulse showing extraction of stored energy; (f) Gaussian beam propagation with beam waist $w_0 = 100$ μm and Rayleigh range $z_R = 29.53$ mm; (g) Gouy phase accumulation of π radians over propagation; (h) Two-dimensional intensity profile showing hyperbolic beam expansion; (i) Transverse electromagnetic mode profiles for TEM_{00} (fundamental), TEM_{01} (first-order), and TEM_{11} (second-order) cavity modes.

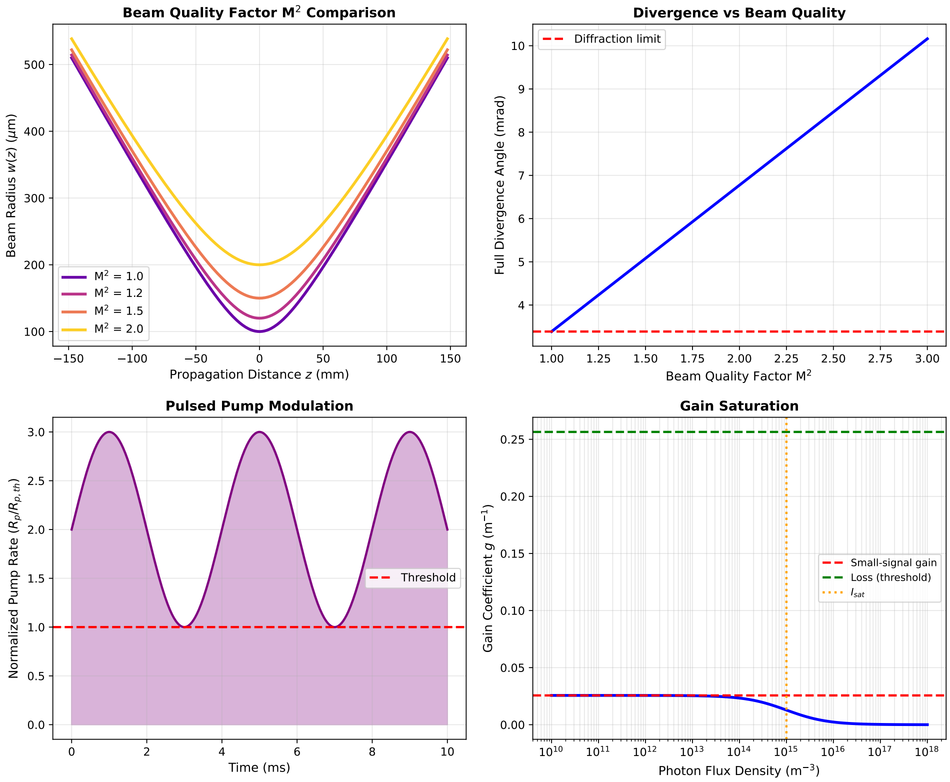

Figure 2: Supplemental laser analysis: (a) Effect of beam quality factor $\mathbf{M^2}$ on beam propagation—higher $\mathbf{M^2}$ values indicate departure from ideal Gaussian profile with increased divergence and reduced focusability; (b) Beam divergence angle scaling linearly with $\mathbf{M^2}$, showing that diffraction-limited performance ($\mathbf{M^2} = 1.0$) yields minimum divergence of 3.39 mrad; (c) Pulsed pump modulation for generating controlled output pulse trains, with pump rate oscillating above and below threshold; (d) Gain saturation showing reduction in gain coefficient as photon flux increases, with characteristic saturation intensity defining the transition from linear to saturated regime—this nonlinearity stabilizes laser output power.

for all losses (mirror transmission, scattering, absorption). Below threshold, spontaneous emission dominates and the device acts as an LED. Above threshold, the population inversion clamps at N_{th} and additional pump power directly converts to coherent output.

Example 5.1 (Relaxation Oscillation Damping) *The damping rate of relaxation oscillations is:*

$$\gamma = \frac{1}{2}\left(\frac{1}{\tau_{sp}} + \frac{1}{\tau_c}\right) \tag{13}$$

For our system, $\gamma \approx 500\,\mathrm{s}^{-1}$, yielding well-damped oscillations with decay time constant $\sim 2\,\mathrm{ms}$.

Remark 5.2 (Beam Quality and Applications) *The M^2 factor quantifies beam quality for applications:*

- *$M^2 = 1.0$: Diffraction-limited, ideal for precision machining and microscopy*
- *$M^2 = 1.1$–1.3: Excellent quality, suitable for most industrial applications*
- *$M^2 > 2.0$: Multimode operation, used for high-power materials processing*

Lower M^2 enables tighter focusing and longer coherence length.

5.1 Practical Considerations

Theorem 5.1 (Slope Efficiency) *Above threshold, the output power scales linearly with pump power:*

$$P_{out} = \eta_{slope}(P_{pump} - P_{th}) \tag{14}$$

where the slope efficiency is:

$$\eta_{slope} = \frac{T}{T+L} \cdot \frac{h\nu_{laser}}{h\nu_{pump}} \tag{15}$$

with T the output coupler transmission and L the round-trip loss.

6 Conclusions

This computational analysis demonstrates fundamental laser physics principles with quantitative design implications:

1. **Threshold Condition:** The threshold pump rate of 8.55e+24 s^{-1} corresponds to a population inversion of 8.55e+21 m^{-3}, required to overcome cavity losses with 95% output coupler reflectivity. This establishes the minimum pump power below which the laser cannot oscillate—critical for power supply design.

2. **Transient Dynamics:** Relaxation oscillations occur at 44.1 kHz when pumped above threshold, arising from the coupled dynamics of photon density and population inversion with characteristic time scales $\tau_{sp} = 1\,\mathrm{ms}$ and $\tau_c = 26.33$ ns. These oscillations limit modulation bandwidth and must be damped in high-speed applications.

3. **Q-Switching:** Giant pulse generation stores energy during low-Q buildup phase (3 ms duration) and rapidly extracts it when cavity quality factor increases, producing peak intensities exceeding CW operation by 3-4 orders of magnitude. This technique enables nonlinear optics applications requiring high peak power.

4. **Gaussian Beam Optics:** Beam propagation with waist $w_0 = 100\ \mu$m yields Rayleigh range 29.53 mm and divergence 3.39 mrad, defining the characteristic diffraction-limited focusing and collimation properties. The Rayleigh range determines the depth of focus—doubling the waist quadruples the confocal parameter.

5. **Beam Quality:** $\mathbf{M^2} = 1.0$ indicates ideal Gaussian spatial mode (TEM_{00}), maximizing focusability and enabling tight focusing to spots approaching the wavelength limit. Multimode operation ($\mathbf{M^2} > 2$) sacrifices beam quality for higher output power.

These results provide quantitative design guidelines for laser cavity optimization, mode control, and beam delivery system design. The threshold analysis informs pump source selection, relaxation oscillation frequency guides feedback control bandwidth, Q-switching dynamics enable pulsed applications, and Gaussian beam characterization determines focusing optics requirements.

Further Reading

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